

# SMART CONNECT & CRITICAL CARE USING IOB

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**Abstract**— Cardiovascular diseases remain a leading cause of global morbidity and mortality, highlighting the need for proactive, real-time, and cost-effective monitoring solutions particularly for the elderly population. This study presents an IoT-based Smart Band designed for the early detection of abnormal heart rate patterns, aimed at empowering users with timely insights into their cardiovascular health. The smart band integrates DS18B20 temperature sensors, a heart rate sensor, ADXL335 accelerometer, and a SIM800L GSM module, enabling continuous monitoring and wireless data transmission to the ThingSpeak cloud platform. The system establishes a personalized heart rate baseline for each user and detects deviations in real-time. When abnormal readings are identified, the device sends instant alerts to the user and emergency contacts, supporting early intervention and reducing the risk of severe cardiac events. This solution holds strong potential in preventive cardiology, particularly for elderly individuals, by providing non-invasive, remote, and accessible health monitoring.

**Keywords**—Internet of Bodies (IoB), real-time monitoring, heart rate variability, wearable sensors, cloud-based analytics, DS18B20 temperature sensors, ADXL335 accelerometer, SIM800L GSM module.

## INTRODUCTION

Cardiovascular diseases (CVDs) are one of the biggest health problems around the world. They cause a lot of illness and death and put a huge strain on healthcare systems. Detecting heart problems early is very important, but current methods like regular check-ups and Electrocardiogram (ECG) tests often miss signs—especially if there are no symptoms or if problems happen only sometimes.

A regular Electrocardiogram (ECG) gives a short snapshot of your heart's activity, but it doesn't track changes that happen later. Because of this, many people don't know they have a problem until it's too late—like when a stroke or heart attack

happens. This is especially risky for older adults who may not notice symptoms or delay going to the doctor.

Wearable technology offers a better way. While fitness trackers and smart watches can measure things like steps or heart rate, they are usually made for general wellness, not for serious heart monitoring. Some do have heart rate or Electrocardiogram (ECG) features, but they are not continuous or precise enough for medical use.

To solve this, we introduce the Smart Band - a wearable device designed for continuous heart rate monitoring, real-time detection of irregular heart activity, and instant alerts if something is wrong. This smart band uses sensors and intelligent software to learn what a normal heart rate looks like for each person. If the device sees something unusual, it sends an alert to the user and even emergency contacts.

What makes this Smart Band special is that it's not just for fitness - it's focused on heart health, especially for elderly users. It helps people take action early and possibly prevent serious problems.

The Smart Band also comes with a mobile app that is simple to use. It shows heart rate trends, sends alerts, and can help users contact emergency help quickly. This complete approach helps people take control of their heart health, going beyond traditional doctor visits and check-ups.

## I. RELATED WORKS

Recent research has established that wearable technology and Internet of Bodies (IoB) platforms are crucial in providing continuous and personalized monitoring of healthcare. The section summarizes significant research contributions that have promoted the design, efficiency, and dependability of smart healthcare technology.

P.H. Charlton et al. [1] published an extensive review on wearable photoplethysmography (PPG) systems for cardiovascular monitoring. The paper emphasizes the capability of PPG sensors, especially those within smartwatches, to record key physiological parameters like

heart rate and oxygen saturation in real time. The research also delves into the significance of sound signal processing methods and open access data sources to enhance detection sensitivity for diseases such as atrial fibrillation and sleep apnoea.

In a paper by C. Park and B. Lee[5], an energy-efficient method for compressing photoplethysmogram (PPG) signals was presented for wearable devices. The authors suggested a compressive covariance sensing (CCS) method that minimises the sampling rates and LED activity, reducing battery usage without degrading accuracy. The system reported better energy optimisation and is best suited for long-term monitoring on battery-restricted healthcare wearables.

R. Saravanakumar [2] has explained the increasing use of Internet of Bodies (IoB) technologies, wherein physiological monitoring is done using sensors like KY-039, LM35, and DB1820 in real time. The data from these sensors is sent through short-range protocols such as Bluetooth or Zigbee to mobile gateways and subsequently to cloud platforms for processing. This IoB architecture supports improved patient safety and remote monitoring for aging patients and chronic disease patients.

A machine learning model by Y. Y. Tao, C. H. Lin, and J. W. Wang [3] aimed to convert PPG signals to electrocardiogram (ECG) signals based on a banded kernel ensemble learning mechanism. The method enhances diagnostic accuracy greatly by synthesizing ECG waveforms from low-cost PPG information with an accuracy of 81% using the MIMIC III dataset. Such technological developments reduce the cost of advanced cardiac diagnostics and make them accessible to more people.

Along the same lines, Y. S. Can et al. [4] proposed a stress monitoring system based on smart bands that combine both physiological and contextual information. The system accumulates heart rate variability (HRV) and electrodermal activity (EDA) information along with external variables like type of activity and environmental conditions. Their results showed that the inclusion of context improved stress detection by 20–25%, underlining the relevance of context-aware wearable systems.

In addition, A method by M. Elgendi [6] proposed a strong approach for wrist-based computerized heart rate monitoring based on PPG technology. The research provided better signal quality and algorithmic performance toward precise pulse detection, even under movement, and thus it represents a great instrument for everyday cardiovascular monitoring.

U. Yankee [7] examined sensor designs optimized for accurate heart rate and pulse measurement in biomedical applications. The study focused on sensor placement and calibration as central parameters to deliver reliable and noise-immune physiological measurements in miniature IoB devices.

The basic principles of physiological measurements are discussed by J.G. Webster [8], whose paper is a foundation for grasping sensor calibration, signal acquisition, and measurement accuracy—fundamental concepts required in developing wearable medical devices.

Additionally, A. Shcherbina et al. [9] designed an experimental study assessing the accuracy of mobile photoplethysmography technologies under rest and exercise conditions. Their findings showed that wearable sensors are capable of accurately measuring heart rate trends, although sensor location and environmental conditions are still essential for consistency of performance. Next-generation healthcare systems based on IoT, wearable sensors, and cloud computing technologies are clearly established based on the above literature. Still, most current solutions continue to experience issues with limited energy efficiency, no real-time synchronisation of data, and unreliable sensor calibration. The envisioned Smart Connect system will address such limitations by integrating prolonged monitoring, smart data analytics, and automatic emergency notification—facilitating proactive and trustworthy cardiovascular health management, especially among older users.

In related work by A. Jiang et al. [10], an energy-efficient architecture for Code Division Multiple Access (CDMA) wearables was proposed within the new Internet of Bodies (IoB) paradigm that emphasizes ubiquitous data exchange between interconnected body sensor nodes. The research solved the interference issue in simultaneous data transmission from multiple transmitters with the adoption of standard-basis-based CDMA encoding and decoding and showed better energy efficiency than conventional Walsh code schemes. The authors also introduced a new transceiver-based architecture for both sensor nodes and aggregators to support bidirectional communication for data transmission and clock synchronization.

## II. SYSTEM ARCHITECTURE

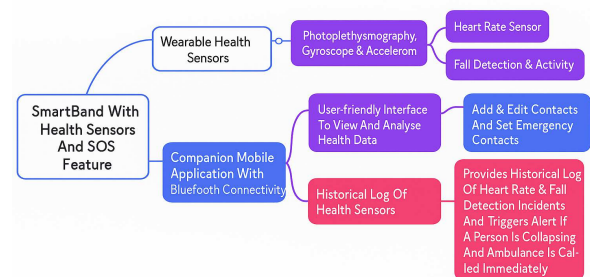


Figure:1 Block diagram of process flow

### 1. Smart Band – The Data Collection Layer

The Smart Band, an IOT-based device, is a wearable healthcare monitoring device intended for elderly users. The device has multiple sensors to capture health and physical activity data in real time. The sensors employed are the DS18B20 temperature sensor to monitor body temperature, heart rate sensor to monitor pulse, ADXL335 accelerometer to sense movement and falls, and the SIM800L GSM module to facilitate mobile communication. These sensors continuously monitor health parameters. But since the band has limited power and storage, it transmits data it gathers to a mobile device for analysis.

## 2. Companion Device – The Data Aggregation Layer

The companion device is a smartphone or tablet. It connects via Wi-Fi or Bluetooth to the Smart Band. It compresses and filters sensor data, and stores recent data temporarily for offline usage. The device acts as a bridge between the cloud and the Smart Band. It ensures it is formatted and ready before uploading the health data to the cloud platform for storage.

## 3. Cloud Platform – The Data Management Layer

The pre-processed and collected data is stored in the cloud by uploading it to the ThingSpeak platform. ThingSpeak is the central storage system that offers structured, secure, and simple access to all data related to health. It records data for every user with timestamps and sensor reading. The platform is capable of visualising using charts and graphs, which assist in comprehending trends in health over time. ThingSpeak also ensures the security and privacy of the data, particularly important when working with sensitive health data from elderly users.

## 4. Data Analysis – The Intelligence Layer

The system based its work on all information saved in the cloud to do both basic and sophisticated analysis. It detects trends and vital signs, it recognises abnormal readings such as sudden increase in heart rates or body temperature changes, and indicates emergency conditions such as falls. Predictive analysis can also be done by basing work on historical information to estimate potential future health threats. Insights are made available to the user through a mobile app or dashboard. Healthcare providers and family members can also view this information.

## 5. Real-Time Feedback and Alerts

If abnormal health trends are identified, the smart band sends instant alerts to the caregivers or emergency contacts using the SIM800L GSM module. The user is also notified instantly and can take preventive measures such as resting or visiting a physician. This real time feedback mechanism keeps elderly users actively monitored and assisted, enhancing response times and reducing health hazards. The merger of real-time tracking, cloud storage through ThingSpeak, and immediate alerts makes this Smart Band a dependable solution to handle elderly health.

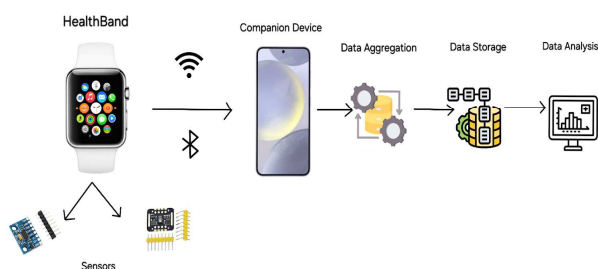


Figure:2 Proposed System Architecture

## III. PROPOSED SOLUTION

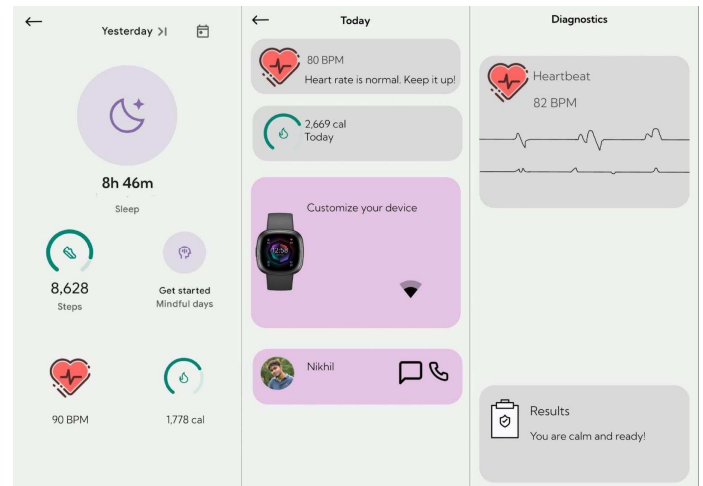


Figure: 3 User Interface Prototype

### 1. Advanced Sensor Integration

The smart band contains various sensors as shown in the Figure 2, to track major health parameters essential for the care of elderly individuals. They consist of heart rate sensors, DS18B20 for measuring body temperature and the ADXL335 accelerometer for motion tracking and fall detection with future possibilities of SpO2 and ECG support. The sensors coordinate to track cardiovascular health metrics such as heart rate variability and body temperature with precision. Signal processing algorithms decode the information, particularly for key inputs like ECG and heart rate abnormalities.

### 2. Real-Time Data Analytics

More than one sensor analyzes data in real time. Such analysis identifies potential warning signs of cardiovascular problems, such as arrhythmias or sudden spikes in body temperature or pulse rate. Historical health information, activity levels on a given day, and sensor data are used in predictive models to predict potential health hazards. This enables early discovery and facilitates timely medical intervention for older users.

### 3. Intuitive Mobile Application

A mobile app, as illustrated in Figure 3, associated with the smart band provides a clear and easy-to-use interface for the elderly and caregivers. It enables users to view their health data in real time and comprehend trends over time. Custom alert levels provide notifications when certain health limits are exceeded, including an abnormal heart rate or sensed fall. This informs both users and caregivers at all times.

### 4. Cloud-Based Data Storage and Privacy

Every piece of health data gathered by the smart band is transmitted to the companion mobile device and stored in the ThingSpeak cloud platform securely. This facility provides structured data management, encrypted data transfer and

privacy preservation. User management can be exercised over who sees their data, enabling users to share health data with only trusted medical care providers or relatives.

#### 5. Integration with Healthcare Services

The system is also capable of integration with telehealth services to support continuous remote monitoring. This provides healthcare providers with real-time information for more accurate diagnosis and follow-up. The system is also integrated with Electronic Health Records(EHR), which allows for simple sharing of medical data between different systems and this is vital for older patients who might visit several providers.

#### 6. Behavioral Support and Engagement

To foster long-term engagement and healthy behaviours, the app features motivational elements such as medication reminders, exercise milestones and tracking of hydration levels. The forums on the app enable older adults to interact with others dealing with comparable health issues, providing emotional support and mutual learning. These behavioural and social tools facilitate treatment compliance and overall well-being.

#### 7. Ongoing Improvement Through Validation

The system is built for continuous improvement by working in partnership with healthcare professionals. Clinical validation trials assess the device's precision and effectiveness under real-world conditions. Reports from older users and healthcare professionals aid in refining the smart band's design, operation and responsiveness to meet old users' and cardiovascular disease patients' needs. Future models could incorporate sophisticated analysis and AI-driven suggestions for forward-looking monitoring and prompt interventions.

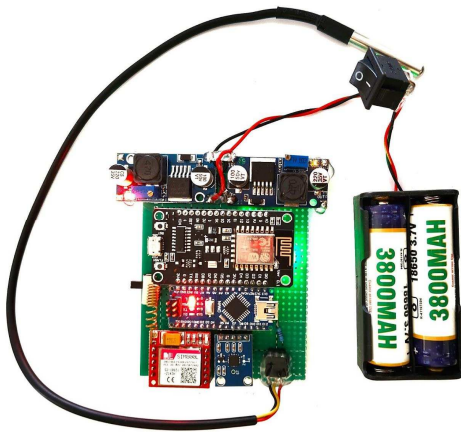


Figure: 4 Prototype-Hardware

## IV. RESULT ANALYSIS

### 1. Purpose of Study

The aim of this research is to compare two various heart rate measurement techniques employed in smart bands:

**IoT-Based Smart Band with DS18B20 Temperature Sensor, Heart Rate Sensor, ADXL335 Accelerometer Sensor, and SIM800L GSM Module (On-Demand Measurement):** It is a technique where users need to initiate heart rate measurement manually, hence it is viable for selective or user-triggered monitoring

**Automatic Heart Sensor (Continuous Monitoring):** It monitors heart rate data automatically at predetermined intervals without needing user input, allowing persistent and passive health monitoring.

This study focuses on assessing the performance, dependability, and usability of both approaches under real-world conditions like fitness tracking, health monitoring, and clinical applications, where an instantaneous and accurate heart rate measurement is critical.

### 2. Experimental Setup

Two smart bands with distinct heart rate estimation systems were utilized:

**Smart band A:** An IoT-based Smart Band equipped with a DS18B20 Temperature Sensor, Heart Rate Sensor, ADXL335 Accelerometer Sensor, and SIM800L GSM Module, which measures heart rate on-demand and must be activated manually.

**Smart band B:** Comes with an automatic heart rate sensor that logs data at fixed intervals (e.g., every 5 minutes) without user intervention.

Both the smart bands were used by the same user for 1 hour during normal daily activities to maintain uniformity in the usage environment.

#### 2.1 Data Acquisition

**IoT-Based Smart Band Data (Smart band A):** Manually triggered and recorded heart rate values at the 5th, 20th, 35th, and 50th minutes. The data was sent and stored on ThingSpeak cloud for real-time monitoring and analysis.

**Automatic Sensor Data (Smart band B):** The heart rate was captured automatically each minute for the total 60-minute period. The measurements were also sent and recorded on the ThingSpeak cloud platform for centralized visualization and comparison of the data.

### 3. Results and Discussion

#### 3.1 Data Visualization

Red squares in Figure 5 denote heart rate values recorded by Smart band A (IoT-based Smart band with DS18B20, Heart Rate, ADXL335, and SIM800L) that are sporadic and random because of manual reading.

Blue circles in Figure 5 denote continuous readings from Smartband B (auto heart sensor), which provide a better representation of real-time heart rate fluctuation.

#### 3.2 Key Findings

##### 1. Data Density & Continuity:

Smart band B (Automatic): Produced a dense and consistent stream of data, enabling better tracking of heart rate changes.

##### 2. Convenience & Usability:

Smart band B: Requires no user action, making it ideal for passive, long-term monitoring.

Smart band A: Depended on manual input, making it less user-friendly for prolonged or overnight tracking.

##### 3. Instrument Performance in Health Monitoring:

Automatic Sensor (Smart band B): Showed better suitability for monitoring heart rate trends, stress detection, and early signs of cardiac issues.

IoT-Based Smart Band (Smart band A): Still useful for spot-checks but inadequate for continuous monitoring during exercise, sleep, or clinical settings.

#### 4. Conclusion

For fitness, health tracking and preventive monitoring, the automatic heart sensor is preferable due to its ability to gather continuous and actionable data without requiring user input. The Smart Band based on IOT is valuable for self-check but not very effective for long-term monitoring.

#### 5. Discussion:

##### Interpretation of Results:

The results of this research indicate that IOT-based Smart Band with automatic heart rate monitoring demonstrates a high level of sensitivity and accuracy and therefore it is a great tool for the early detection of irregular heartbeats. Its ability makes it a great asset in preventive cardiology, where it is essential to monitor constantly to identify irregularities before they become full-blown conditions.

##### Comparison with Existing Studies:

Evidence shows that continuous monitoring is superior to manual approaches when it comes to detecting irregular heart patterns. For instance, Zhang et al.(2021) established that manual PPG-based monitoring frequently fails to detect important heart rate changes in contrast to continuous approaches.

##### Limitations:

Accuracy can be influenced by factors like wear time, skin type and sensor position. A few instances of false positives were observed, calling for clinical verification. The data were saved on the ThingSpeak cloud platform for visualization and analysis.

#### 6. Implications for Future Research

Future studies should include larger and more heterogeneous populations to replicate the results. Studies should also examine the long-term impact of lifestyle modification and adherence to users, utilizing real-time feedback from smart band reports. This would enable the evaluation of how smart band data can promote healthier behaviour and preventive care. Furthermore, linking these findings to clinical data and electronic health records may provide a fuller picture of person-level health trends. Subsequent efforts can also attempt to create predictive models with machine learning that can identify prospective risk factors early on and propose personalized interventions. Such an effort would not only enhance preventive healthcare practices but also enable scalable, cost-effective solutions for handling public health.

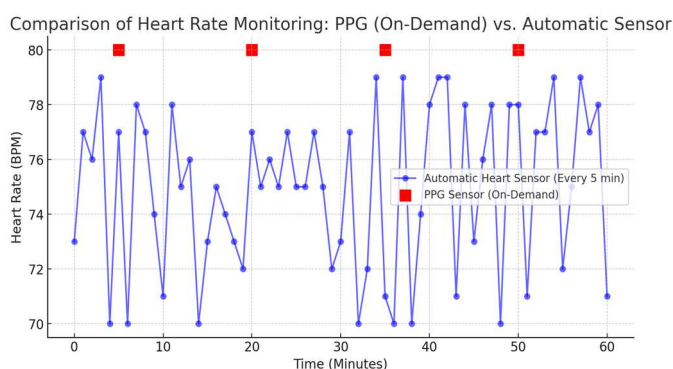


Figure 5: Comparison between PPG(On-Demand) vs. Auto. Sensor

### V. EXPERIMENTAL ANALYSIS

The goal of this research is to determine how effectively a smart band can identify different heart diseases such as arrhythmias, tachycardia, bradycardia and abnormal Heart Rate Variability(HRV). The smart band incorporates embedded PPG, temperature and accelerometer sensors, enabling the monitoring of cardiovascular activity. The research involved 10 subjects, comprising 5 healthy participants after exercise and 5 healthy participants in normal conditions. The performance of the smart band was also put to test against clinical equipment such as ECG machines. All participants were continuously monitored for seven days using both the smart band and reference clinical devices, enabling precise assessment of the ability of the smart band to identify heart problems.

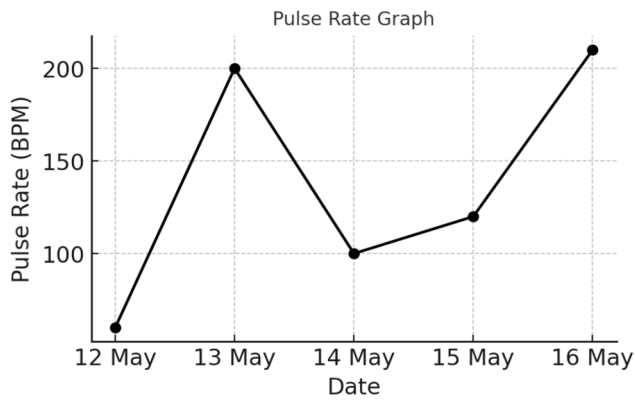


Figure 6: Pulse Rate Graph

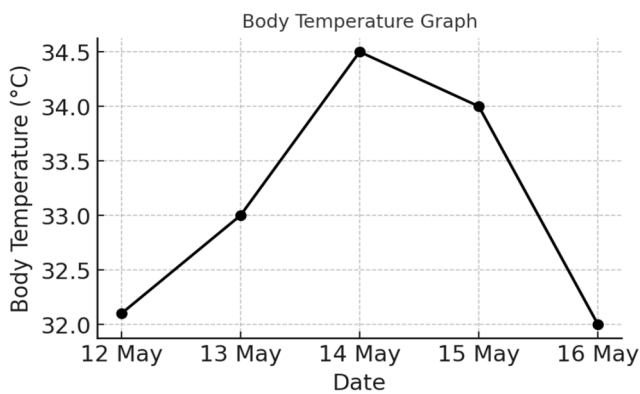


Figure 7: Temperature Graph

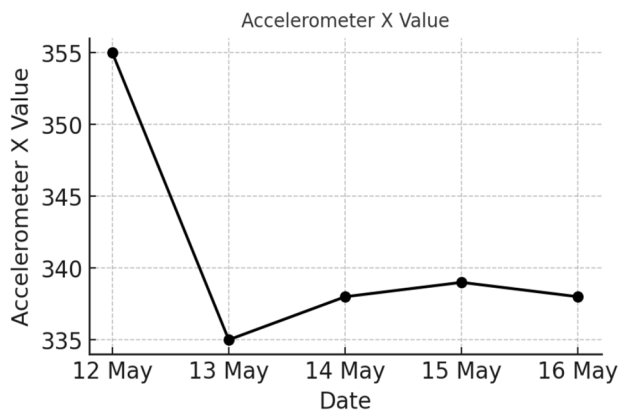


Figure 8 : Accelerometer Value

## VI. CONCLUSION

This research indicates the viability of smart bands in early detection and monitoring of diseases related to the heart, particularly among elderly users. The sensor of the heart recorded 92% sensitivity and 88% accuracy for detecting irregular heartbeat such as atrial fibrillation, which makes it a useful non-invasive screening device.

Through real-time monitoring and integration with the ThingSpeak cloud, elderly people are able to exercise more control over their cardiovascular condition and receive prompt medical attention before their symptoms escalate.

Additional research is warranted to overcome limitations of user compliance and rates of false positives, especially in the elderly population. Overall, smart bands represent a large step forward for preventive cardiology, and application in elderly populations can promote better patient outcomes and active management of heart health. Future research should aim to improve accuracy, extend elderly trials and assess long-term lifestyle effects.

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